

What is a high fiber diet?

There is no recognized definition of what constitutes a high fiber diet. Intakes of dietary fiber in different populations internationally vary widely from less than 20 g to more than 80 g per day. The types of foods contributing fiber also vary; in some countries cereals contribute the most fiber, in others leafy or root vegetables predominate. Vegetables have the highest fiber content per Kcal, and in most populations with fiber intakes over 50 g, vegetables contribute over 50% of total fiber intake. In rural Uganda, where the fiber hypothesis was first developed by Burkitt and Trowell, vegetables contribute over 90% of fiber intake. An experimental diet, the "Simian" diet, has been developed to mimic as closely as possible using human foods, the diet consumed by our simian ancestors the great apes. It is also similar to the Ugandan diet in containing large amounts of vegetables and 50 g fiber/1000 Kcal. Though nutritionally adequate, this diet is very bulky and not a suitable model for general recommendations. Dietary guidelines are that fat intake should be < 30% of energy, with a fiber intake of 20-35 g/d. These recommendations are inconsistent with a high fiber diet because, for people consuming more than about 2400 Kcal, low fiber choices for fruits and grains must be selected to keep dietary fiber intake within the range of 20-35 g. In a 30% fat, 1800 Kcal omnivorous diet, selection of wholemeal bread and whole fruit, results in a fiber intake over 35 g/d, and for and 1800 Kcal vegetarian diet, with substitution of modest amounts of peanut butter and beans for meats, dietary fiber intake goes up to 45 g/d. Thus, if it is desirable to promote the use of unrefined foods, the recommended dietary fiber intake should be a minimum of 15-20 g/1000 Kcal.